Lecture Notes on Data Engineering
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The Same Size Distribution of Data Based on Unsupervised Clustering Algorithms

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Abstract. It is known that dividing data into groups based on certain rules not only helps to separate meaning from the data but also increases the efficiency of the large-volume data processing process. The same size distribution of data is especially effective in approaches such as distributed computing or parallel processing. The main reason for this is that dividing the data into as equal clusters as possible allows achieving the highest efficiency results in these approaches. But the distribution of data in the same size based on the human factor is a complicated process due to the impossibility of pre-planning the data contained in the data flow and the size of the data. In such a situation, unsupervised clustering algorithms are one of the main solutions to the problem of uniform data distribution. In this research work, hierarchical, K-means, Bisecting K-means, and DBSCAN unsupervised clustering algorithms are analyzed in order to solve the given problem, and the results of research on equal clustering of different groups of data are presented using them. Moreover, a new unsupervised equal-size clustering algorithm for equal clustering of data is proposed. During the research, the efficiency indicator of this algorithm for dividing data into equal groups is compared with the indicators of existing algorithms, and the conclusions of the experimental results are presented.

Keywords: Unsupervised clustering algorithms · hierarchical clustering · K-means · Bisecting K-means · DBSCAN · new equal clustering algorithm

1 Introduction

Nowadays, data analysis, which is, identifying hidden contents from a data set, and separating them into useful, reliable, and interrelated data groups is one of the topics of modern research [1]. In the modern digital age, the increase in the volume of data flow increases the relevance of this research topic. Because as the flow of information increases, people's natural data analysis capabilities are limited, and the process of extracting meaningful information from big data becomes more complicated [2–4]. In such a situation, the most effective solution for finding hidden patterns in data is unsupervised learning based on data clustering approaches according to certain criteria. Unsupervised clustering is an approach to grouping data by grouping or separating those with similar patterns, and those with different patterns from the data without human intervention [5].

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In these clustering approaches, the grouping of data mainly focuses on characteristics such as similarity and uniqueness of the data, and usually ignores or does not consider the number of elements in the groups to be very important. But in real life, there are such problems that not only the similarity and difference of data are important in solving these problems, but also the equality of the number of elements in clusters is an important factor [6, 7]. For example, when storing and processing data using parallel and distributed computing mechanisms. In these mechanisms, the more evenly the data is distributed to the parts of the system, the higher the efficiency [8–10]. Because in this case, all distributed functions within the system perform the same number of tasks, and all parts of the system complete their tasks at the same time. In other words, they do not wait for each other's tasks to be completed. This is the most important indicator of achieving the highest efficiency in the work process. Therefore, achieving equal size distribution based on unsupervised clustering algorithms was taken as the major research objective.

In order to solve the problem presented during the research, the capabilities of hierarchical, K-means, Bisecting K-means, and DBSCAN existing unsupervised clustering algorithms to divide data into clusters of equal size are analyzed. However, due to the limited capabilities of existing algorithms, a new unsupervised clustering algorithm is proposed to achieve uniform distribution of data. At the end of the study, the efficiency indicator of the proposed algorithm for dividing data into groups of equal size is compared with the indicators of existing algorithms, and opinions and comments are given on the results of the experiment.

2 Methodology

In this research work, first of all, the literature related to the solution of the problem was analyzed. It was found that most of the literature covered only the content of unsupervised clustering algorithms [13–19]. Only in some literatures, studies related to the size of clusters have been conducted [11, 12]. However, in these literatures, the problem of clustering of the same size has not been considered.

During the research, the selection of the characteristics of the data was also considered in order to divide the data into clusters. In this process, the ASCII codes of the characters in the data were used and the following features were extracted. The ASCII code of the first character of the data, the number of vowels and consonants, the sum of their ASCII codes, the number of total characters of the data, and the sum of its ASCII codes.

At the next stage of the research, the data were distributed using existing unsupervised clustering algorithms based on the features extracted above, and the sizes of the clusters were determined. The research process showed that existing unsupervised clustering algorithms are not always effective in solving the given problem. Therefore, a new vector module square-based clustering algorithm was proposed during the research. These steps are described in the following sections of this article.

3 Opportunities of Existing Unsupervised Clustering Algorithms to Distribute Data into Clusters of Equal Size

3.1 Hierarchical Clustering

Hierarchical clustering is one of the widely used unsupervised algorithms for clustering data into groups. Hierarchical clustering mainly uses two approaches. These are agglomerative hierarchical clustering and divisive hierarchical clustering [11, 13]. In both approaches, the clustering result can be illustrated using a dendrogram (Fig. 1).

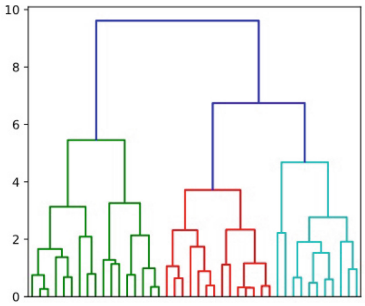

Fig. 1. Representation of the result of hierarchical clustering using a dendrogram

In hierarchical clustering algorithms, grouping is done after the complete cluster hierarchy is built. That is, after all the elements in the collection are formed as branches of a single tree, the branches of the tree corresponding to the required number of clusters are separated. This process can be visualized using the dendrograms shown in Fig. 2.

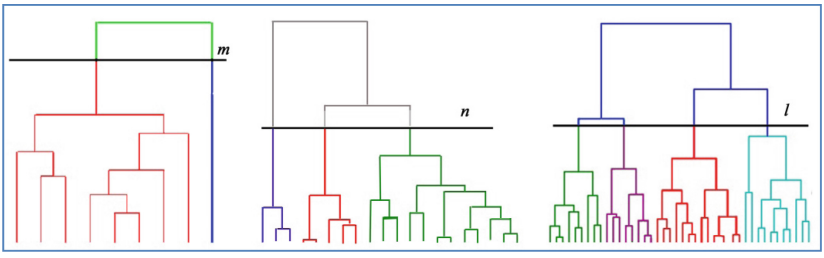

Fig. 2. Line m divides the first dendrogram into 2 clusters, line n divides the second dendrogram into 3 clusters, line l divides the third dendrogram into 4 clusters

In Fig. 2, line m in the first dendrogram divides objects into 2 clusters, line n in the second dendrogram divides objects into 3 clusters, and line l in the third dendrogram divides objects into 4 clusters. As can be seen, only for the objects in the third dendrogram, the size of the clusters is almost equal. It can be concluded that it is not always possible to divide the data into groups of (almost) the same size by means of hierarchical clustering, which usually depends on the state of the objects to be clustered.

3.2 K-means Clustering

The K-means clustering algorithm is a center-based grouping method, and the objects in the set are clustered according to which of the central points known as centroids are close [14–17]. The number of centroids is the same as the number of clusters requested by the user. Centroids are initially chosen arbitrarily. Therefore, clustering will not be perfect in the initial state. Minimization of the sum of squares of the errors of the objects in the clusters relative to the centroids is used to make the clusters perfect (1).

$$S = \sum_{i=1}^{k} \sum_{x_j \in C_i} dist(c_i, x_j)^2 \tag{1}$$

where k- is the number of clusters, C_i- is the cluster, c_i is the centroid coordinate of the i cluster, x_j is the j element coordinate in the C_i cluster. It is known that S reaches its minimum value when the derivative of S with respect to c_i is equal to 0 (2).

$$\frac{\partial S}{\partial c_i} = \frac{\partial S}{\partial c_i} \sum_{i=1}^{k} \sum_{x_j \in C_i} dist(c_i, x_j)^2 = 0 \tag{2}$$

Based on this formula, the optimal value of the centroid of the n^{th} cluster can be found as follows.

$$\frac{\partial S}{\partial c_n} = \frac{\partial S}{\partial c_n} \sum_{i=1}^{k} \sum_{x_j \in C_n} (c_n - x_j)^2 = 0 \tag{3}$$

If the number of objects in the n^{th} cluster is m, then the following solution comes from Eq. 3

$$2 \cdot \sum_{j=1}^{m} c_n = 2 \cdot \sum_{j=1}^{m} x_j \quad \Rightarrow \quad m \cdot c_n = \sum_{j=1}^{m} x_j \quad \Rightarrow \quad c_n = \frac{\sum_{j=1}^{m} x_j}{m} \tag{4}$$

The conclusion from formula 4 is that the centroids reach optimal coordinates when they are equal to the average value of the coordinates of the objects in the cluster.

In the K-means algorithm, objects are divided into clusters of equal size compared to hierarchical clustering. This is especially evident when objects are located at the same density in the coordinate system.

In Fig. 3a, the objects are divided into 3 clusters, and the size of each cluster is almost the same. But if these objects are divided into 2 or 4 clusters, the sizes of the clusters will be very different from each other. In Fig. 3b, it can be seen that two clusters are large in size, and the other two are small in size. Figure 3c shows a perfect set of objects, i.e., a set of objects that are relatively evenly distributed over space. When this set is divided into an arbitrary number of clusters, clusters of almost the same size are formed.

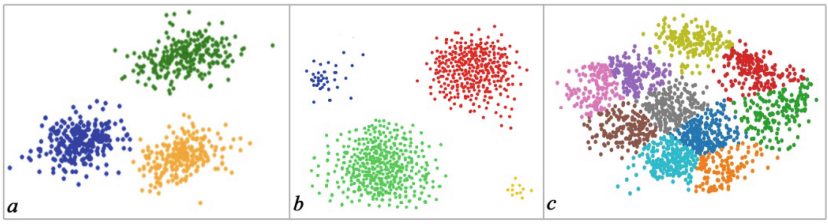

Fig. 3. The result of object clustering using K-means

3.3 Bisecting K-means Clustering

The Bisecting K-means algorithm is an improved interpretation of the K-means algorithm, which is more efficient than K-means in splitting into clusters of equal size. Because the main idea in Bisecting K-means is to create a new cluster by dividing a cluster with a large size or a large sum of squared errors [18]. This process is shown in Fig. 4 can be seen.

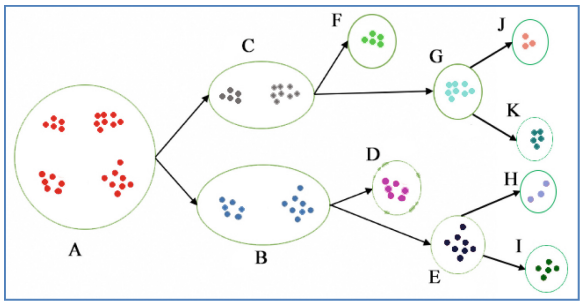

Fig. 4. Clustering of objects using the Bisecting K-means algorithm

In Fig. 4, the first set A is given. Collection A is divided into clusters B and C based on the Bisecting K-means algorithm. When the number of clusters is equal to 3, cluster B is divided into two since there are more objects in cluster B than in cluster C. Then three clusters C, D, and E are formed. When the number of clusters is equal to 4, cluster C is divided into two since cluster C has more objects than clusters D and E. After that and 4 clusters named D, E, F, G are formed. As the number of clusters increases, the same processes are continued.

3.4 DBSCAN Clustering

DBSCAN is a density-based clustering algorithm that combines objects that are adjacent to each other at a distance of *max_dist* (maximum distance) into one cluster [19, 20]. The second important parameter of DBSCAN is *min_points* (minimum number of objects). It means the minimum number of objects forming a cluster. That is, if the number of objects located at *max_dist* distance is less than *min_points*, these objects do not belong to any cluster and they are considered as noise.

This clustering method is more powerful than other clustering methods. In other words, this algorithm is the most effective approach to identifying hidden meanings. But this algorithm is ineffective in solving the problem of equal clustering. Because in the DBSCAN clustering method, it is impossible to specify the number of clusters in advance by the user. When the number of clusters changes depending on *max_dist* and *min_points*, it is impossible to say exactly how many clusters will be formed in advance. In addition, some information contained in the data to be clustered may not belong to any clusters, that is, they may be considered as noise. Taking into account these shortcomings, it can be said that it is impossible to use the DBSCAN algorithm in solving the problem of equal-sized clustering of data streams.

4 Clustering Algorithm Based on Square of Vector Modulus

In this research, a new clustering algorithm is proposed in order to solve the problem of distributing the data flow into clusters of equal size. This clustering algorithm is based on the construction of vectors directed from the coordinate origin to the feature coordinates of objects. It is known that any *n*-dimensional vector $\vec{a}$ with coordinates $(x_1; x_2; x_3; \ldots; x_n)$ has a length of a certain positive number and it is found by formula 5.

$$|\vec{a}| = \sqrt{x_1^2 + x_2^2 + x_3^2 + \ldots + x_n^2} \tag{5}$$

This proposed algorithm is based on dividing the squares of vector lengths into groups based on certain rules. After the objects are converted to numerical values, clustering based on length values makes it even possible to separate two objects that are very close to each other into other clusters. This makes it possible to equalize the sizes of clusters during clustering.

The proposed clustering algorithm based on the square of the vector modulus is implemented by performing the following steps:

Step 1. Read the data (objects);
Step 2. Transfer the specific characteristics of each data to vector coordinates in digital form;
Step 3. Calculate the square of the modules of the resulting vectors;
Step 4. Sort the squares of the modules of the vectors in ascending order;
Step 5. Write the squares of the modules of the vectors sorted in ascending order into the array A;
Step 6. Input k which number of clusters;
Step 7. Divide array A into k equal parts;
Step 8. Determine the boundaries of the pieces;
Step 9. Find the boundaries of the cluster through the boundaries of the slices;
Step 10. Separate into clusters the data using defined thresholds.

As a result of the execution of this algorithm, the elements of *k* clusters separated have their own geometric meaning. The elements of the cluster 1 consist of the points of a circle with a radius *R*, the elements of clusters 2, 3,... k − 1 are the points of the rings

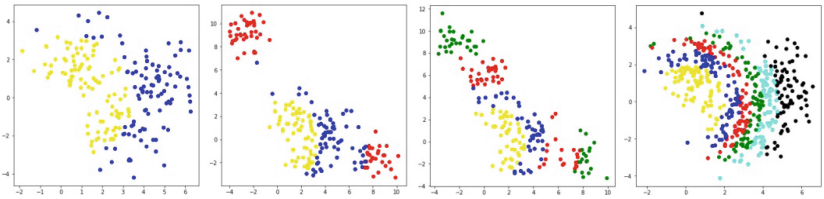

Fig. 5. Graphical representation of clustering based on the square of the vector modulus

with thicknesses d_2, d_3, $\ldots d_{k-1}$, and the elements of the cluster k are the points lying outside the circle with the radius R_1. Here $R_1 = R + d_1 + d_2 + \ldots + d_{k-1}$ equality will be appropriate. The results of the proposed algorithm can be seen in Fig. 5.

It can be seen from Fig. 5 that in the clustering based on the square of the vector module, the points that are very close to each other also belong to two other clusters. On the contrary, the points located far from each other belong to one cluster. This feature is another indicator that distinguishes the proposed algorithm from other clustering algorithms. But the main feature of this algorithm is that the sizes of the clusters formed as a result of clustering are almost equal.

5 Results

In order to check the ability of clustering algorithms to divide into clusters of equal size, 3 different data sets containing 1000 elements were used. Each data set was divided into 2, 3, 4 clusters. The elements within these clusters were enumerated and the results compared.

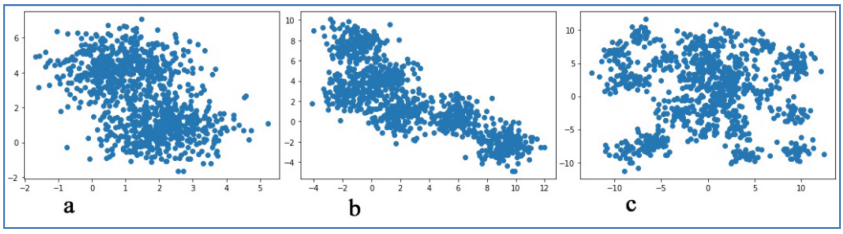

Fig. 6. 3 different data sets used in the experiments

3 types of data shown in Fig. 6 were used in the experiments. The data used in the experiment was generated by the random function and does not have any special meaning. But these data were created using the make_blobs data generation function of the sklearn.datasets library of the python programming language, and data were generated around 2, 6, and 30 central points, respectively.

Table 1. The results of the distribution of a, b, and c data into 2 groups using clustering

The name of the algorithm	Clusters №	Clusters size		
		a	b	c
Hierarchical clustering	1	2	2	999
	2	998	998	1
K-means clustering	1	521	364	436
	2	479	636	564
Bisecting K-means clustering	1	508	364	596
	2	492	636	404
The proposed clustering	1	500	500	500
	2	500	500	500

Table 2. The results of the distribution of a, b, and c data into 3 groups using clustering

The name of the algorithm	Clusters №	Clusters size		
		a	b	c
Hierarchical clustering	1	2	2	997
	2	997	4	2
	3	1	994	1
K-means clustering	1	272	430	380
	2	257	329	339
	3	471	241	281
Bisecting K-means clustering	1	250	213	468
	2	259	423	211
	3	491	364	321
The proposed clustering	1	333	333	333
	2	333	333	333
	3	334	334	334

Table 3. The results of the distribution of a, b, and c data into 4 groups using clustering

The name of the algorithm	Clusters №	Clusters size		
		a	b	c
Hierarchical clustering	1	2	2	997
	2	6	4	1
	3	991	993	1
	4	1	1	1
K-means clustering	1	245	264	215
	2	246	232	146
	3	269	186	223
	4	240	318	416
Bisecting K-means clustering	1	216	364	365
	2	275	272	233
	3	250	151	202
	4	259	213	200
The proposed clustering	1	250	250	250
	2	250	250	250
	3	250	250	250
	4	250	250	250

As can be concluded from Tables 1, 2 and 3, the proposed vector length-based clustering algorithm recorded a good result in terms of the characteristic of equal-sized clustering of most data for 3 types of data. This situation can be seen in all 2, 3, and 4 clusters. The worst indicator belongs to the Hierarchical clustering algorithm, in all cases, the sizes of the clusters were very different from each other. In all cases, a large part of the data belonged to a certain cluster, and a small part belonged to other clusters. K-means and Bisecting K-means algorithms performed relatively well as expected, but in no case did they perform better than the proposed approach.

The result of the proposed algorithm for solving the given problem can be seen in Fig. 7.

Figure 7 shows the distribution of the data a, b, and c into 4 clusters based on the proposed algorithm, and the 4 clusters are shown in 4 different colors.

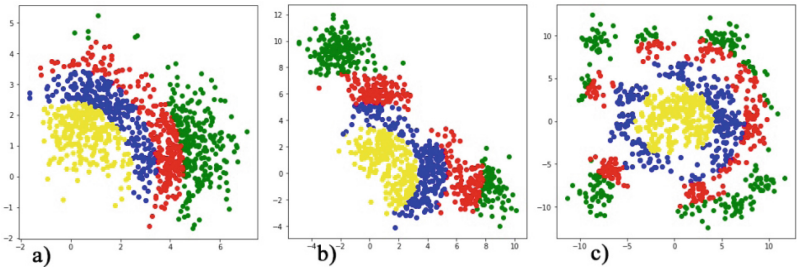

Fig. 7. The result of the distribution of the data set a, b, and c into 4 clusters based on the proposed algorithm

6 Conclusion

In conclusion, this research work was devoted to studying the possibility of clustering algorithms to divide data into clusters of equal size. For this purpose, this possibility of existing unsupervised clustering algorithms Hierarchical, K-means, Bisecting K-means and DBSCAN clustering algorithms was studied. The results of the research showed that existing unsupervised clustering algorithms have strong capabilities in identifying hidden content in data, but limited capabilities in dividing clusters of equal size. In order to overcome these shortcomings, a clustering algorithm based on the square of the length of the proposed vector was proposed and the steps of the algorithm were presented. The research results showed that this algorithm is very effective in solving the given problem.

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